

can also keep 1 GB if you have 512 MB or more of RAM and want to conserve disk space.

4. If the disk management tool is not already open, right click *My Computer*, click *Manage*→*Disk Management*.
5. On the second hard disk (Disk 1), create a primary partition 1.5 times the size of your RAM (or 1 GB—see the above point); plus about 15 to 20 per cent of that space, assign a drive letter and format it (preferably with NTFS). Note that only Windows needs the extra 15-20 per cent of free disk space. Don't reserve any extra amount for Linux on the first disk. Close the disk management tool when done.
6. Right click *My Computer*, click *Properties*→*Advanced*. Under *Performance* click *Settings*→*Advanced*. Under *Virtual memory* click *Change*.
7. Select the first partition (C:), click *No paging file*, then click *Set*.
8. Select the new partition (drive letter) you created in Step 5, then click *Custom size*.
9. If you allocated 1.5 times the amount of your RAM, type that amount in megabytes (1 GB = 1024 MB) in both the minimum and maximum boxes, then click *Set*. If you allocated 1 GB, type 1024 in both boxes, then click *Set*. This sets up your virtual memory for Windows. Make sure not to set the whole partition space (which includes the additional 15-20 per cent space) here. The extra free space is required by Windows to perform optimally.
10. Click *OK*, then *OK* again to close the *Performance Options* box.
11. Click the *System Restore* tab, then select the partition (drive letter) you set the virtual memory in.
12. Click *Settings*.. then select "Turn off System Restore on this drive" and click *OK*. Confirm that you want to turn off System Restore on the drive.
13. Click *OK* to close the 'System Properties' box.
14. In your Linux installation program, select the option for custom (or expert) partitioning at the beginning of the partitioning phase (most Linux installers provide such an option, but if yours doesn't, you can use the *fdisk* or *cfdisk* program; (*c*)*fdisk* coverage is out of the scope of this article so you can get help from their manual and info pages on a working/live Linux system).
15. Select the first hard disk and create a primary partition in the space you had reserved for Linux swap at the end of the first disk. Make the partition use the full reserved (available) free space. Set it as the swap partition. This creates the virtual memory for Linux.
16. Select the second hard disk and create primary or extended partitions as you want for your Linux files after the existing Windows partition. The existing Windows partition on the second disk contains the Windows virtual memory that was set up previously.
17. Proceed with your Linux installation. Don't set the option to automatically mount the Windows virtual memory partition on the second hard disk. If your installer turns it on by itself, make sure to turn it off.

Some notes

The exact performance gains you enjoy will vary depending on the amount of RAM you have (the less the RAM, the more the virtual memory used), the number of applications you use at a time and the load they place on the system memory, and other similar factors. However, even if not perceptible, there will be some difference, which will be better than using the default virtual memory set-up. You can check the amount of virtual memory being used, by using the *Task Manager* in Windows and the *System Monitor* in Linux.

A maximum of four primary partitions is allowed on a hard disk. More than four partitions can be created by using logical partitions within an extended partition. If you want to use logical partitions for your Windows files, fill the first hard disk to the end with the extended partition, create your logical partitions and make sure to keep the Linux swap space reserved within the extended partition. Then, in Step 15, create a logical partition for the swap space instead of a primary partition. Note, however, that this is not the optimal set-up for virtual memory.

If you have more than two hard disks and want to install more than two operating systems on them, you can jumble up the virtual memory partitions for them in a similar manner. However, note that cross swapping can be implemented best if you have as many operating systems as hard disks and install each OS on a different hard disk. Also ensure that you keep the first partition on the first hard disk for Windows.

If you use IDE hard disks, the primary master is the first disk, the primary slave the second, the secondary master the third, and the secondary slave the fourth. However, for best performance, you should make your hard disks masters on separate IDE channels (e.g., primary master and secondary master). For SATA hard disks, the disks are ordered according to the SATA ports on which they are connected.

Windows does not recognise/read Linux and UNIX partitions; so it is safe to place the swap partition for Linux/UNIX on the hard disk on which Windows is installed. However, do not try to manipulate any Linux/UNIX swap or other partitions using the Windows disk management tool.

You can allot partitions for the virtual memory of an OS on more than one hard disk with one partition on each hard disk. This will improve performance further. However, don't allot a virtual memory partition for an OS on the hard disk on which the OS itself is installed, else you will defeat the purpose of cross swapping.

If you use more than one Linux distribution, you can reuse the same virtual memory partition(s) for all of them. 

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